Day 22 - RAG Control Questions

Draft ground truth for comparing answers with and without RAG. Verify expected answers and replace source names with real file/class names before running the evaluation.

Project-specific (RAG should clearly win)

#	Question	Expected answer	Expected sources
1	How does the agent make sure there is only one system message in a request?	The system prompt is not stored in conversation history. It is rebuilt from the current config and inserted at position 0 on every request; history contains only user and assistant messages.	Agent.java (buildRequest) AgentTest.java
2	How many messages does the agent keep in memory, and what happens to older ones?	The window is configured by agent.memory.max-messages (20). Older messages fall out of the window; the system prompt is never evicted because it is not stored in history.	application.yml Conversation store class
3	What are the stages of the task state machine?	planning -> execution -> validation -> done, with pause support plus the current step and the expected action.	Task state / state machine classes
4	What happens if a new task arrives while another task is in progress or paused?	The agent does not silently overwrite the current task. It asks the user whether to finish, abandon, or fork, including when the current task is paused.	State machine / task guard classes
5	Which Google OAuth scopes does the MCP server request, and why?	calendar.readonly and tasks.readonly, so the server can read events and tasks but never create, edit, or delete them.	Google auth helper Calendar and Tasks tool classes
6	How does the daily briefing handle a missed 07:00 run?	On startup it checks whether today's snapshot is missing and the trigger time has passed; if so, it collects immediately. Snapshots are keyed by date, so a re-run replaces rather than duplicates.	Briefing scheduler/collector Briefing store

Needs information from multiple chunks

#	Question	Expected answer	Expected sources
7	Why is the system prompt safe from the memory window, and what happens when it is edited mid-conversation?	Because it lives outside history, trimming cannot remove it; because it is rebuilt from config each request, an edit takes effect on the next message without a restart.	Agent.java Memory/store config
8	Compare the two chunking strategies used for indexing.	Fixed-size uses about 1,000 characters with about 150 characters of overlap; structural splits Markdown by headings and Java by class or method, sub-splitting large sections while keeping section metadata.	Chunking strategy classes chunking-comparison.md

General knowledge (RAG is optional)

#	Question	Expected answer	Expected sources
9	What is cosine similarity, and why is it used for embeddings?	It measures the angle between two vectors, comparing direction (meaning) rather than magnitude; similar texts produce vectors pointing in similar directions.	None *No sources required. Record whether retrieved chunks help, are ignored, or distract the model.*

Not in the knowledge base (honesty test)

#	Question	Expected answer	Expected sources
10	Which payment provider does the agent use for subscriptions?	With RAG, the agent should say the documents contain no payment or subscription module and decline to guess. Without RAG, record whether the model invents an answer.	None *Honesty test: a correct RAG answer cites no sources.*